The Mechanism Layer in `cdm` v0.3

Novelty and feasibility of how an option works

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Abstract

This document describes the mechanism layer added to `cdm` in version 0.3: what it computes, how it changes decisions, how to use it, and the evidence for and against it. The layer describes each option by the functions it must perform and the working principle it uses for each. Novelty is computed from that description instead of from a judged score, an unproven working principle is sent to a prototype test instead of being vetoed, and the outcome of each test updates a belief that is shared by every option using the same principle. Full derivations are in the paper, *Unknown Is Not Risky* (v0.3.0), §5 and Appendix A.

1 Purpose

Version 0.2 decided which options were novel with one ordinal question to the judge ("how different is the approach from every precedent?"). This had two consequences.

1. The score depends on how an option is written. On Jev, describing the same mechanism with promotional wording raised the score by 0.058 on a $[0, 1]$ scale.
2. The score does not identify a mechanism. A check of one option therefore produced no information about any other option, even one built on the same idea.

Engineering design addresses both by describing a solution as a set of *functions*, each performed by a *working principle* (the morphological chart of Zwicky; the function structures of Pahl and Beitz). The mechanism layer adopts that description. Novelty and feasibility become properties of principles, and principles are shared across options and across decisions.

2 Concepts

Function. A result the solution must achieve, for example "collect the monthly fee". Each function has a weight w_f and a catalogue of known working principles.

Working principle. A way of achieving a function, for example "cash collected on a visit" or "mobile money". A principle outside the catalogue is labelled NEW.

Record. The precedents: for each, which principle it used for each function. The record defines what counts as familiar.

Classification $\pi_{i,f}$. The judge's distribution over the catalogue plus NEW for option i and function f, asked as a Choice question over all cyclic orderings so that position bias cancels.

Risk estimate $r_{i,f}$. The judge's probability that the way option i performs f fails in this setting (a Noul question).

3 Quantities

quantity	definition	meaning
principle novelty	$N_f(p) = 1 - \min(1, c_{f,p}/m_f)$; $N_f(\text{NEW}) = 1$	1 if no precedent used p for f (default $m_f = 1$)
mechanism novelty	$\nu_i = \sum_f \bar{w}_f \sum_p \pi_{i,f}(p) N_f(p)$	weighted probability of an unused principle
combination novelty	$\kappa_i = 1 - \sum_{r \in R^*} \prod_f \pi_{i,f}(r_f)$	probability the combination is in no precedent
reliability belief	$\bar{\theta}_{f,p}$ (see below)	probability that p performs f here
mechanism risk	$q_i^{\text{mech}} = 1 - \prod_f \sum_p \pi_{i,f}(p) \bar{\theta}_{i,f,p}$	probability the mechanism fails
learning value	$\Lambda_i = \sum_f \sum_p \pi_{i,f}(p)\, \bar{\theta}(1-\bar{\theta})/(n+1)$	what a test of the option would teach

The reliability belief for a principle comes from the first applicable source:

1. *Settled*: probability 1, for a principle proven reliable (below). Options that use only settled principles are deployed without a check.
2. *Evidence*: a Beta(a, b) recorded from tests and from precedents' history, shared by all options that use the principle.
3. *Opinion*: Beta$(2(1 - r_{i,f}) + c_{f,p},\ 2r_{i,f})$, the judge's estimate held as two pseudo-observations, plus one success for each precedent that used the principle.

When a principle counts as proven

A principle is settled only if its recorded evidence satisfies

$$\mu - \sigma\sqrt{(1-\alpha)/\alpha} \ \geq\ \theta_{\min} \qquad (\theta_{\min} = 0.95,\ \alpha = 0.05),$$

where μ and σ are the mean and standard deviation of its recorded Beta. By Cantelli's inequality this guarantees $P(\theta < \theta_{\min}) \leq \alpha$ for any belief with that mean and variance. In practice a principle needs on the order of a hundred recorded uses with almost no failures. Use by precedents without a recorded history proves nothing: record their track record with `morphology.evidence[(f, p)] = Beta(successes + 1, failures + 1)`. The v0.3.0 rule, which settled any principle a precedent had used, is available as `Morphology(settle="precedent")`; `settle="never"` checks everything.

4 Effect on decisions

The layer changes the policy in three places and leaves everything else unchanged.

1. **Innovation slot eligibility.** ν_i replaces the judged novelty, with its own gate $\nu_{\min} = \frac{1}{2}\min_f \bar{w}_f$ (0.25 for two equally weighted functions): the midpoint between a repeated mechanism ($\nu = 0$) and one unused principle in the lightest function. A reworded precedent has $\nu = 0$ and cannot take the slot. Override with `Policy(min_mechanism_novelty=...)`.
2. **Prototype gate.** q_i^{mech} is added as a hard constraint. An option built on an untested principle has q^{mech} near 1/2 and is placed in the *verify* state: it is tested, with a fallback to the status quo if the test fails. Evidence of failure raises q^{mech} until the option is rejected.
3. **Optional learning bonus.** With `Policy(learning_weight=w)`, the innovation slot ranks options by $\text{EI}_i + w\Lambda_i$. The default is $w = 0$.

Commit, portfolio, $P(\text{best})$ and EVPI are computed as before. The axioms P1–P8 of the policy continue to hold, because novelty and the learning bonus enter only the innovation slot.

5 The value of a test across decisions

For a single decision, only the mean of the belief about a violation matters (paper, Proposition 1). Across decisions the spread matters. For a principle with belief $\text{Beta}(a, b)$, $n = a + b$, $\bar{\theta} = a/n$, and a later decision that gains $\alpha(\mathbb{E}[\theta] - t)$ when it acts:

$$\text{VOI}(t) \;\le\; \alpha\,\frac{\bar{\theta}(1-\bar{\theta})}{n+1},$$

with equality at $t = \bar{\theta}$, and $\text{VOI}(t) = 0$ outside an interval of width $1/(n+1)$. A principle with no evidence ($\text{Beta}(1, 1)$) is worth $101/3 \approx 34$ times as much to test as one whose reliability is known to be one half ($\text{Beta}(50, 50)$).

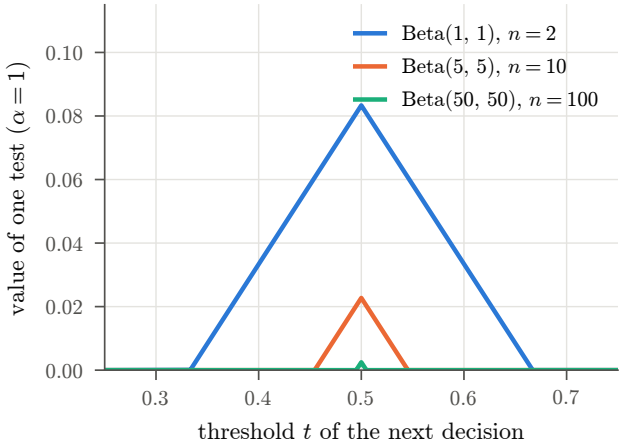


Figure 1: Value of one test for a later decision with threshold t, for three beliefs with mean 0.5.

6 Usage

Related decisions should run through a `Programme`, which keeps the evidence about each principle and credits every reported test to the principles the option uses.

```
from cdm import (Beta, Function, JevJudge, Morphology, Policy, Problem,
                 Programme, judge_problem)

reach = Function("reach", "find shop owners and convince them to sign up", {
    "door": "Staff visit shops in person",
    "referral": "Owners are rewarded for bringing in other owners"})
pay = Function("pay", "collect the monthly fee from shops", {
    "cash": "Cash collected during a visit",
    "momo": "Payment from the owner's mobile-money wallet"})
morph = Morphology([reach, pay], record=[{"reach": "door", "pay": "cash"}])
# 412 past visits, 3 failed: enough to prove door-to-door reliable
morph.evidence[("reach", "door")] = Beta(410, 4)
```

```
problem = Problem(goal=..., context=..., options=[...],
                  dimensions=[...], morphology=morph)
programme = Programme(morph, Policy(check_sensitivity=0.9,
                                    check_specificity=0.95))
judgments = await judge_problem(problem, [JevJudge()])
decision = programme.decide(problem, judgments)
decision.by_id("momo").tests        # [("pay", "momo")]

# a prototype of mobile money failed; every option using it
# is re-priced at the next decision
programme.report("momo", passed={"pay": False})
programme.save("evidence.json")     # carries over between sessions
```

Each `OptionStats` reports `novelty` (ν), `combination` (κ), `learning` (Λ) and, among `checks`, `"mechanism"` when a prototype test is required, and `tests`, the principles that test has to cover. A test is credited to an option's principles per function where results are per function, and not at all for a function whose classification is less than 0.5 certain. Shah's rarity metric is available by passing `familiar_after=None` to `Morphology`.

7 Evidence

Six experiments test the layer; the paper (§6.7–6.12) gives the details. Every number below is checked against the result files by `scripts/verify_figures.py`.

experiment	design	main result
wording test (Jev)	28 mechanisms, plain and hype	principle named in 111 of 112 cases; hype moves judged novelty +0.058, ν +0.005
disguise test (Jev)	104 options: plain, hype, jargon, mundane	jargon on repeated mechanisms: judged +0.096, ν +0.043; no wording changed an admission at the default gates
sequential benchmark	simulated, 24–48 related decisions, 300 worlds per setting	learning lowers regret in every setting; late commitments to unreliable principles 26% → 7%
robustness sweep	one assumption varied at a time, 200 worlds	learning pays unless classification is near chance; robust to interactions and catalogue size
Jev in the loop	Jev's real mechanism judgments, simulated outcomes, 1200 worlds	regret halved against v0.2 (−0.037); ablation below
settling rules	as above, precedents of mixed reliability	evidence rule: lower regret and harm (6.3% → 3.7%) than settling on precedent use

Which part of the layer is useful

The Jev-in-the-loop ablation splits the gain over v0.2 (regret 0.074):

- **Novelty measure** (ν instead of the judged score): no gain (−0.0003). On Jev, the judged novelty already separates repeated from new mechanisms; ν is better calibrated (error 0.054 against 0.128) and moves less with wording, but this did not change decisions.

- **Precedents as evidence** (the settled rule): 0.026 $[0.023, 0.028]$, with harm rising from 3.3% to 5.3%, because precedents' principles are deployed without a check.
- **Learning per principle**: 0.012 $[0.010, 0.014]$ with the settled rule, 0.014 $[0.011, 0.016]$ without it, where it also lowers harm to 2.6%.
- **Innovation pilot**: 0.0004 $[-0.0006, 0.0013]$, not measurable.

The layer is useful because naming principles lets evidence attach to them. These results set the defaults: learning per principle through a `Programme`, and a principle settled only when its recorded history proves it reliable. The innovation pilot stays small or off.

Which precedents to treat as proven

With precedents of mixed reliability (half with $\theta = 0.995$, half between 0.7 and 0.9) and a 300-use history each, the v0.3.0 rule settled every precedent principle: more than half of its commitments relied on an unreliable one, and harm was 6.3%. The evidence rule never settled an unreliable principle, settled the reliable ones, and lowered regret by 0.0043 $[0.0027, 0.0059]$ and harm to 3.7%. With only 30 past uses it settled nothing, which still cut harm to 3.4% at no significant cost in regret. Recording no history gave the lowest harm (3.1%) but the highest regret.

When learning stops paying

In the robustness sweep the regret reduction from learning was 0.0117 with perfect classification, 0.0076 at 95%, 0.0033 at 70% and 0.0010 (not significant) at 50%. With half of all pairs of untested principles failing together, it was still 0.0064. Jev classified 95–100% of cases correctly.

8 Limits

- The catalogue of principles is written by people. A poor catalogue produces poor novelty.
- A principle filed under NEW accumulates no shared evidence until it is catalogued.
- Principles are assumed not to interact, and classifications to be independent across functions. κ indicates where the first assumption matters most.
- The settled rule treats a precedent in the same setting as proof that a principle works there.
- The Jev tests use templated options. Outcomes in all sequential experiments are simulated and share the model's structure; none establishes better decisions on real programmes.
- The default gate on ν was introduced after the disguise test showed that the earlier gate (0.5) admitted only 56% of new mechanisms; it is derived from the structure of ν, not fitted.